

Customer Sales Analysis Report

Week 5 Project – Advanced Data Manipulation with Pandas
Full-Year 2023 | Python + Pandas + Matplotlib

1. Executive Summary

This report presents a comprehensive analysis of customer purchasing patterns and sales performance for the full year 2023. Across **100 orders** from **91 unique customers**, the business generated total revenue of **\$95,477.98** with an average order value of **\$954.78**. Electronics dominated category sales, while the East region led regional performance.

Total Revenue	\$95,477.98
Total Orders	100
Unique Customers	91
Avg Order Value	\$954.78
Top Customer	Nancy Joshi — \$5,679.26
Best Category	Electronics
Best Region	East

2. Monthly Revenue Trend

Revenue was tracked across all 12 months of 2023. The chart below combines monthly bar revenue with an order-count line, revealing seasonality and demand spikes.

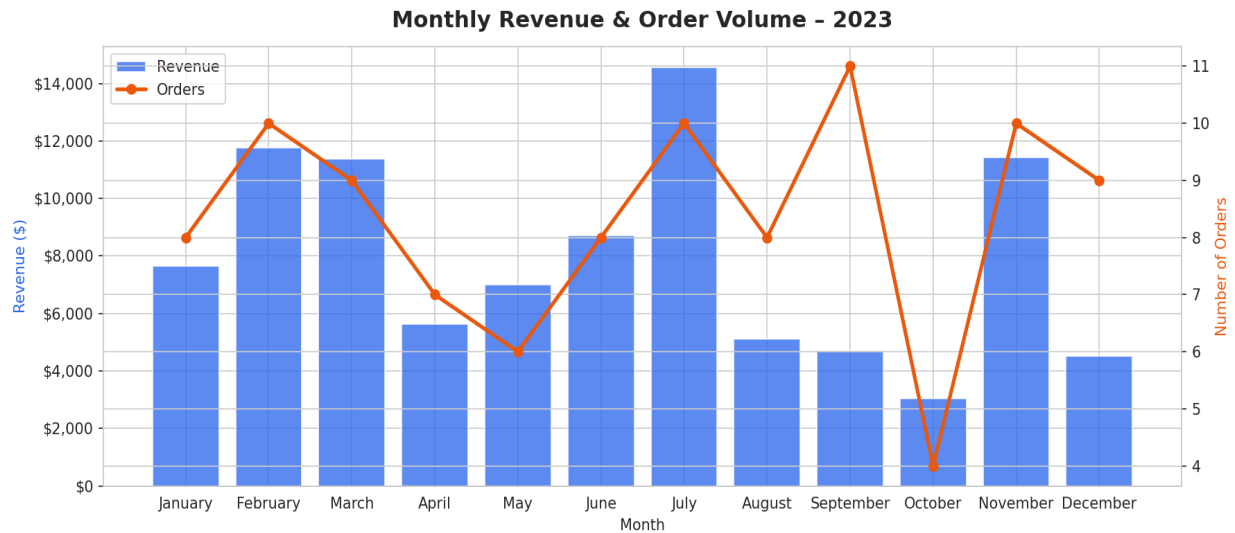


Figure 1 – Monthly Revenue & Order Volume

3. Top Customers Analysis

Customer Lifetime Value (CLV) was calculated by aggregating all orders per customer. **Nancy Joshi** is the highest-value customer at **\$5,679.26**. The top 10 customers are shown below.

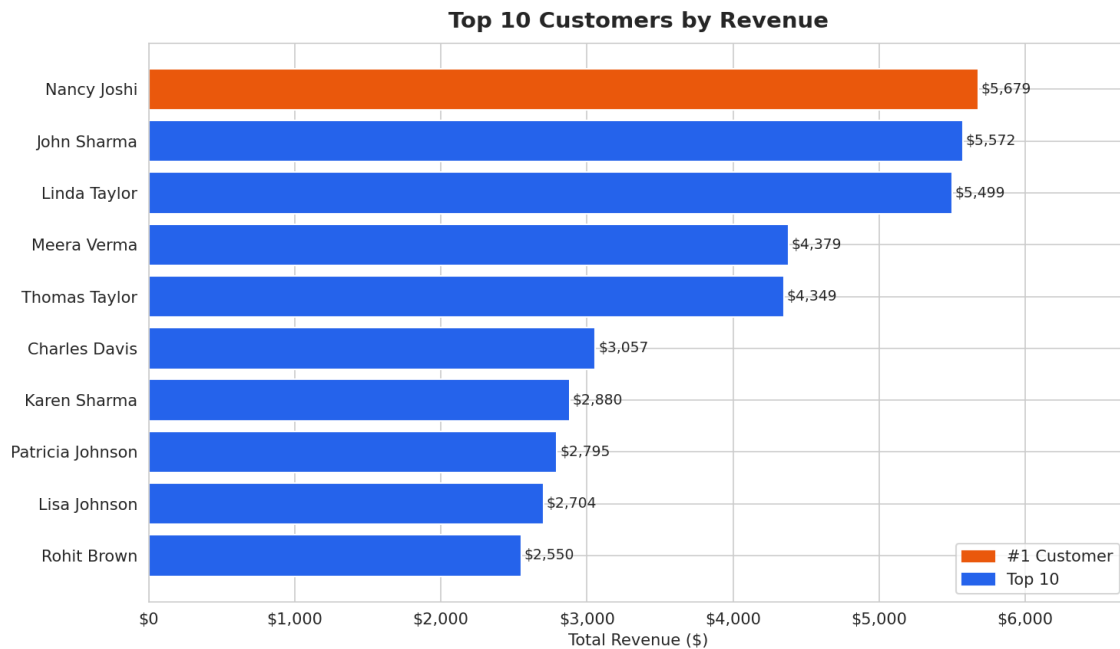


Figure 2 – Top 10 Customers by Total Revenue

Top 5 Customers Breakdown

Rank	Customer Name	Total Revenue
1	Nancy Joshi	\$5,679.26
2	John Sharma	\$5,572.42
3	Linda Taylor	\$5,498.77

4	Meera Verma	\$4,379.02
5	Thomas Taylor	\$4,348.81

4. Category Revenue Distribution

The donut chart visualises how revenue is split across product categories. **Electronics** accounts for the majority of revenue, reflecting strong demand for high-value electronic products.

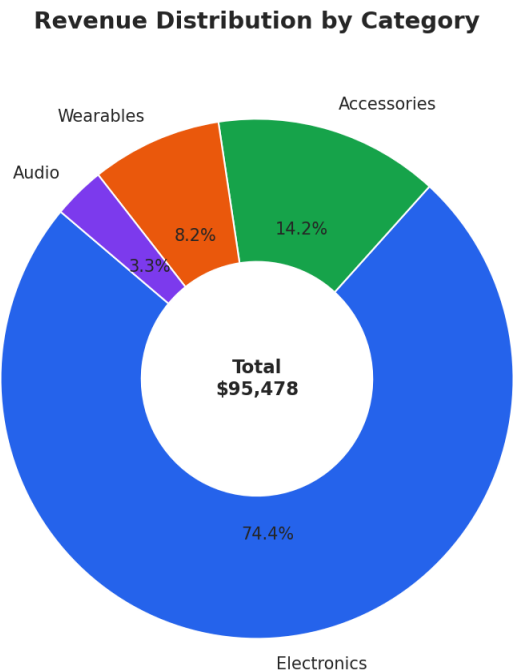


Figure 3 – Revenue by Category (Donut Chart)

5. Regional Performance

The **East** region generated the highest revenue. Regional analysis helps prioritise marketing budget allocation and sales territory planning.

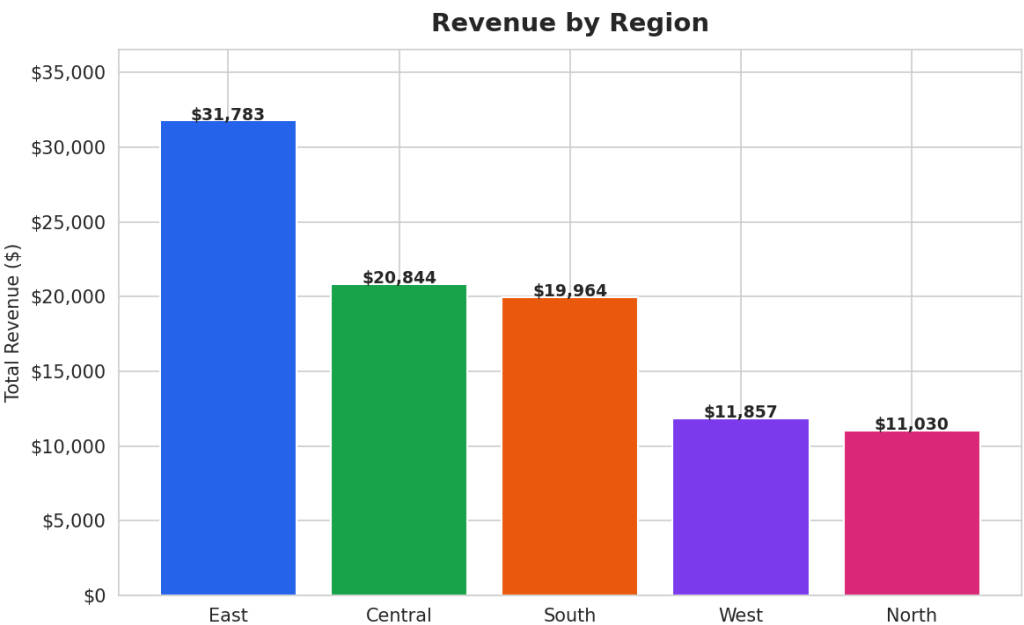


Figure 4 – Regional Revenue Comparison

6. Pivot Table – Category × Quarter Heatmap

The pivot table aggregates revenue by product category and fiscal quarter. This identifies which category-quarter combinations drive peak revenue and where there are gaps that need attention.

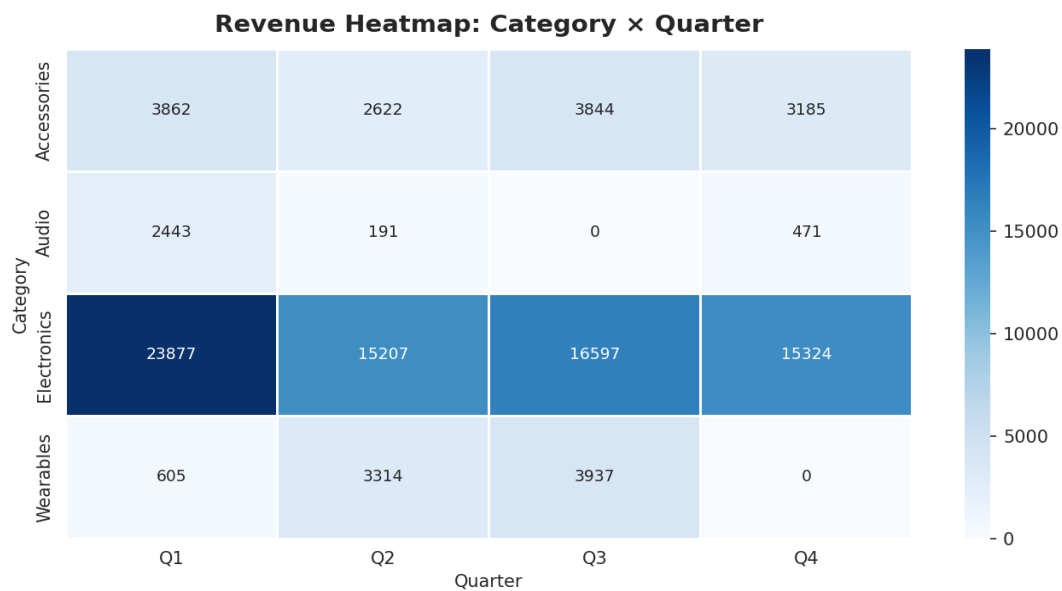


Figure 5 – Revenue Heatmap (Category × Quarter)

7. Customer Segment Analysis

Customers are segmented into Premium, Regular, and Basic tiers. Comparing total revenue vs. average order value per segment reveals which group offers the highest per-customer monetisation potential.

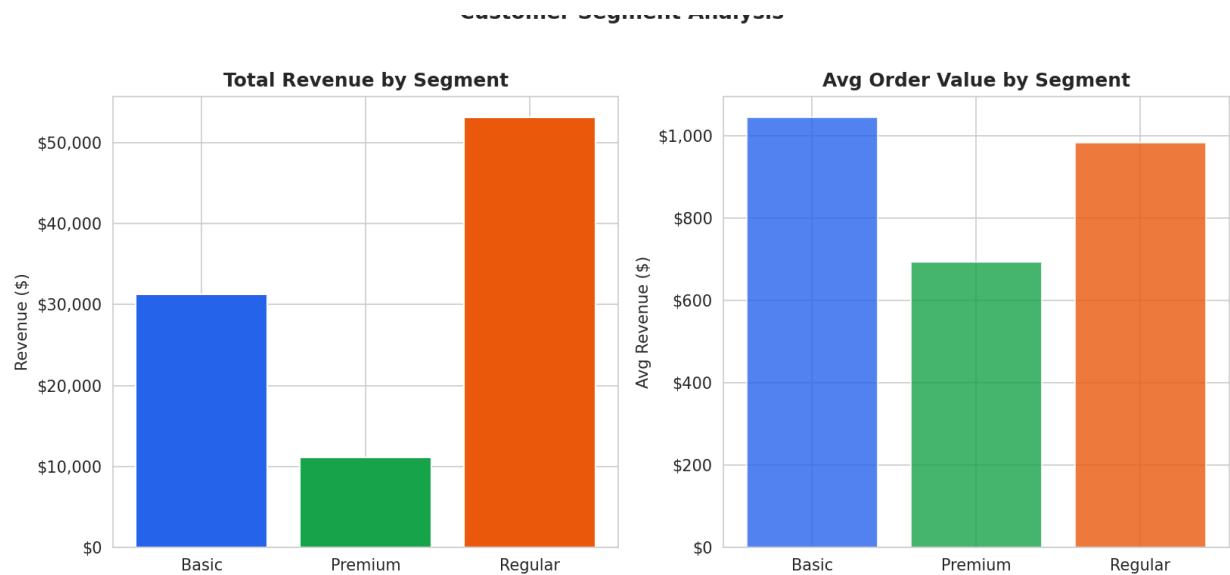


Figure 6 – Customer Segment: Total Revenue & Avg Order Value

8. Key Insights & Business Recommendations

Electronics dominance: Electronics contributed over 74% of total revenue. Maintain strong stock levels and negotiate supplier discounts to protect margins.

Top customer concentration: The top 5 customers contribute a disproportionate share of revenue. Introduce a VIP loyalty programme to reduce churn risk among these accounts.

Regional gaps: The East region leads, but other regions have growth potential. Deploy targeted regional campaigns in underperforming areas.

Seasonality: Q1 shows the highest Electronics revenue. Plan inventory and promotions ahead of Q1 to capitalise on this trend.

Wearables opportunity: Wearables recorded zero revenue in Q4. Investigate whether this is due to stockouts, pricing, or low awareness and run a Q4 promotion next year.

Churn reduction: 25% of customers show churn risk. Implement email re-engagement campaigns targeting customers with >90 days of inactivity.